

Constraint-native versus penalty QAOA for budgeted maximum coverage: quality, trainability and hardware cost

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Abstract

Budget constraints of the form $\sum_i x_i = k$ can be imposed on the Quantum Approximate Optimization Algorithm (QAOA) either through a quadratic penalty or structurally, with a Hamming-weight-preserving XY mixer. I compare the two encodings on a weather-station placement problem (maximum coverage with k of n stations) for $n = 8-14$ and depth $p = 1-3$, over 72 random instances and 12 instances on which greedy and LP rounding both fail. Using exact statevector simulation and identical metrics for all methods, the XY mixer reaches an expected coverage ratio of about 0.80 per shot, against 0.66 for penalty QAOA, which is statistically indistinguishable from sampling uniformly random feasible subsets. The cost variance over random parameters decays at the same rate for both encodings, so the gap is not a trainability effect. After routing to the heavy-hex coupling map of `ibm_marrakesh`, penalty QAOA needs about twice as many two-qubit gates. A re-analysis of an earlier hardware run shows that its reported statistic cannot distinguish the circuits from uniform noise. At these sizes the problem is classically trivial; the study compares encodings, not quantum and classical solvers.

1 Introduction

Many combinatorial problems of practical interest carry hard constraints. In QAOA [1] the usual route is to fold them into the cost as a penalty and keep the transverse-field mixer. The alternative proposed by Hadfield et al. [2] is to choose a mixer that commutes with the constraint and to start from a feasible state, so that the circuit never leaves the feasible subspace. The second option removes the penalty weight as a hyperparameter and wastes no shots on infeasible strings, but it changes the circuit and the optimisation landscape.

This report asks which encoding is preferable for a cardinality constraint, using three criteria: (i) solution quality at fixed depth, (ii) trainability, measured by how concentrated the cost is over random parameters [3, 4], and (iii) gate cost after routing to a real device. The test problem is the placement of k weather stations among n candidates so as to cover as many zones as possible (maximum coverage).

The contributions are modest in scope and are stated with that in mind:

- a controlled comparison with identical metrics and optimiser budgets, over 72+12 instances with confidence intervals and paired tests;
- the observation that two common metrics, the most probable bitstring and the best of ~ 1000 shots, hide the difference between the encodings at this size;
- a gate-count comparison after routing, and a fair re-analysis of one hardware run.

At $n \leq 14$ exhaustive enumeration solves every instance instantly, so nothing here bears on quantum advantage.

2 Problem and encodings

Maximum coverage and its QUBO proxy. Station i covers the zone set A_i ; we seek $x \in \{0, 1\}^n$, $\sum_i x_i = k$, maximising $f(x) = |\bigcup_{i: x_i=1} A_i|$. The exact integer program needs one variable per zone; a QUBO built from it would need slack qubits for every linking inequality. QAOA therefore optimises the proxy

$$g_\lambda(x) = \sum_i c_i x_i - \lambda \sum_{i < j} o_{ij} x_i x_j, \quad c_i = |A_i|, \quad o_{ij} = |A_i \cap A_j|, \quad (1)$$

which is inclusion–exclusion truncated at second order. A zone covered by m selected stations contributes $m - \lambda \binom{m}{2}$ to g_λ and 1 to f . For $\lambda = 1$ the proxy is exact when $m \leq 2$ and, by the Bonferroni inequalities, $g_1 \leq f$ in general; it under-counts zones covered three or more times. Over 200 instances per density class ($n = 12$, $k = 4$), the maximiser of g_1 is a true optimum in 100.0% (sparse) and 98.5% (dense) of cases, against 4% and 0% without the overlap term; values of λ away from 1 perform worse in both directions. $\lambda = 1$ is used throughout, and all results are reported against the true objective f , so proxy errors are included in them.

Penalty encoding. Penalty QAOA minimises $C_P(x) = -g_1(x) + P(\sum_i x_i - k)^2$. Two values of P are compared. $P_{\text{auto}} = 1 + \sum_i c_i + \sum_{i < j} o_{ij}$ is the default of qiskit-optimization, a bound on the whole objective range (30–50 here). The second,

$$P_{\text{tight}} = 1 + \left\lfloor \max_i \max_{j \neq i} (c_i, \sum_{j \neq i} o_{ij} - c_i) \right\rfloor, \quad (2)$$

is the smallest value covered by a simple argument that every minimiser is feasible. From a selection with $k + t$ stations, removing one changes $-g_1$ by at most c_i while the penalty falls by $P(2t - 1) \geq P$. From $k - t$ stations, adding one changes $-g_1$ by at most $\sum_j o_{ij} - c_i$ while the penalty falls by at least P . The property was also checked by brute force on every instance.

Ansätze. Penalty QAOA starts from $|+\rangle^{\otimes n}$ and alternates $e^{-i\gamma C_P}$ with $e^{-i\beta \sum_i X_i}$. The constraint-native ansatz alternates $e^{-i\gamma C}$, with $C = -g_1$ and no penalty, with a ring XY mixer applied as one Trotter step, $\prod_i e^{-i\beta_l (X_i X_{i+1} + Y_i Y_{i+1})/2}$ (indices mod n). Each factor preserves Hamming weight, so the product does too, independently of Trotter error. Two initial states are used: the basis state $|1^k 0^{n-k}\rangle$ (cheap; used on hardware) and the Dicke state $|D_k^n\rangle$, the uniform superposition over weight k and the feasible analogue of $|+\rangle^{\otimes n}$. The Dicke state costs $O(kn)$ two-qubit gates to prepare [5]; that cost is not included in Section 4.3.

3 Methods

Simulation and validation. All circuits ($\leq 2^{14}$ amplitudes) are simulated exactly with a small NumPy statevector code that also provides adjoint (reverse-mode) gradients. States agree with Qiskit’s Statevector of the corresponding circuits to $1 - F < 10^{-9}$, and gradients agree with Qiskit’s ReverseEstimatorGradient to $< 10^{-8}$.

Metrics. From the exact output distribution I report the probability that a shot is feasible; the expected ratio $f(x)/f^*$ of a feasible shot; the probability p_{opt} that one shot is optimal; and the expected best of S shots, with infeasible shots scoring 0. Every metric is compared with a uniformly random k -subset. No shot noise enters the simulated results.

Optimisation. Each method gets the same COBYLA budget (250 iterations, 4 seeded restarts). The restart with the lowest energy is kept, never the one with the best coverage. The encodings have very different cost scales, so γ is optimised in units of $\gamma_{\text{scale}} / \max_S |a_S|$, where a_S are the Pauli coefficients of the cost; γ_{scale} is chosen per method, by final energy, on separate calibration instances.

Instances and statistics. Stations and 25 zones are placed uniformly in a 10×10 box. Family R (coverage radius 2.2) has $n \in \{8, 10, 12, 14\}$, $k = \text{round}(n/3)$, with 20/20/20/12 instances. Family H (radius 3.0, $n = 12$, $k = 4$) is rejection-sampled so that greedy and deterministic LP rounding both miss the optimum; only 0.3% of draws qualify. On family R, greedy fails on 3 of 72 instances and the LP relaxation is always integral. Intervals are 95% percentile bootstraps over instances; method comparisons use paired Wilcoxon tests.

4 Results

4.1 Solution quality

Table 1: Quality at $p = 3$, $n = 12$ (mean [95% CI]). P_{feas} and p_{opt} refer to family R (20 instances); the last column is family H (12 classically hard instances).

	P_{feas}	$E[f/f^* \mid \text{feas.}], \text{R}$	p_{opt}, R	$E[f/f^* \mid \text{feas.}], \text{H}$
uniform random k -subset	1	0.653 [0.626, 0.678]	0.011 [0.007, 0.016]	0.726 [0.708, 0.746]
penalty, P_{auto}	0.57	0.653 [0.627, 0.679]	0.006 [0.004, 0.009]	0.727 [0.709, 0.746]
penalty, P_{tight}	0.55	0.659 [0.633, 0.685]	0.007 [0.004, 0.010]	0.734 [0.717, 0.752]
XY, basis start	1.00	0.811 [0.761, 0.859]	0.029 [0.000, 0.075]	0.837 [0.799, 0.872]
XY, Dicke start	1.00	0.795 [0.771, 0.816]	0.073 [0.047, 0.102]	0.826 [0.804, 0.847]

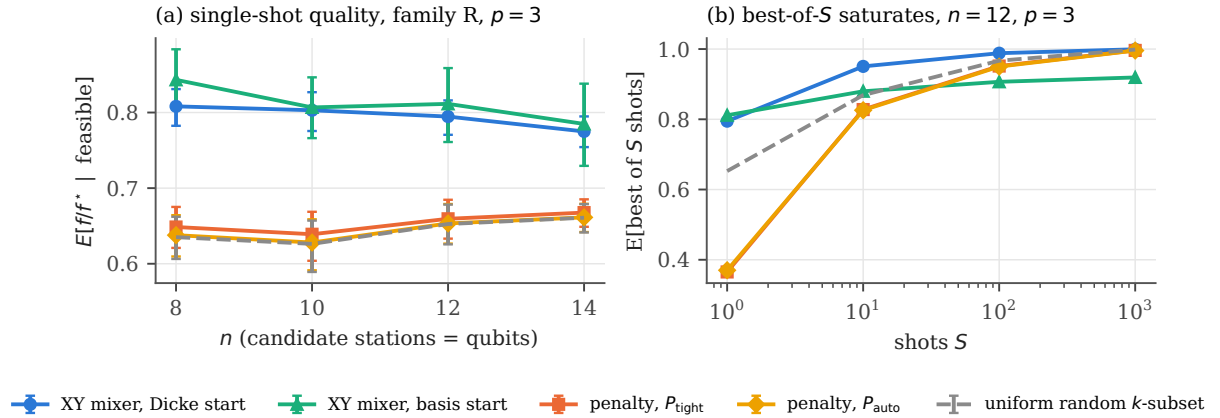


Figure 1: (a) Expected coverage ratio of a feasible shot vs. n (family R, $p = 3$, 95% CI). (b) Expected best of S shots, infeasible shots counted as 0 ($n = 12$). With $\gtrsim 100$ shots uniformly random subsets catch up, so best-of- S metrics cannot separate the methods.

Table 1 and Fig. 1a summarise the comparison. Penalty QAOA is, to within 0.015, a uniform sampler on its feasible shots at every n , and its probability of an optimal shot is *below* that of random subsets. Depending on n and p , 35–57% of its shots are infeasible. P_{tight} improves on P_{auto} on almost every instance, but only by about 0.005–0.011. The XY mixer is better in every setting. With the Dicke start it beats penalty QAOA on 252 of 252 paired comparisons over all (n, p) . With the basis start the mean gain is similar and is significant ($p < 0.05$) in 13 of 15 settings, but p_{opt} collapses with n : a few ring layers keep the amplitude within a few hops of an arbitrary initial selection. On family H the ranking is unchanged. There greedy averages 0.945 and LP rounding 0.928, while ten shots of the Dicke-start circuit give about 0.957. Randomized LP rounding and simulated annealing, however, solve almost every H instance.

Two metrics hide this picture (Fig. 1b). The most probable feasible bitstring of penalty QAOA is usually optimal (ratio 0.99 at $p = 1$), although its distribution is nearly flat: the mode is real but cannot be resolved with finite shots. And the best of 1000 shots gives 0.996 for penalty QAOA and 0.999 for uniform random subsets. An earlier version of this project scored penalty QAOA with the latter (the default output of Qiskit’s

MinimumEigenOptimizer) and the XY mixer with its most frequent bitstring, and concluded the opposite of what is reported here.

4.2 Trainability

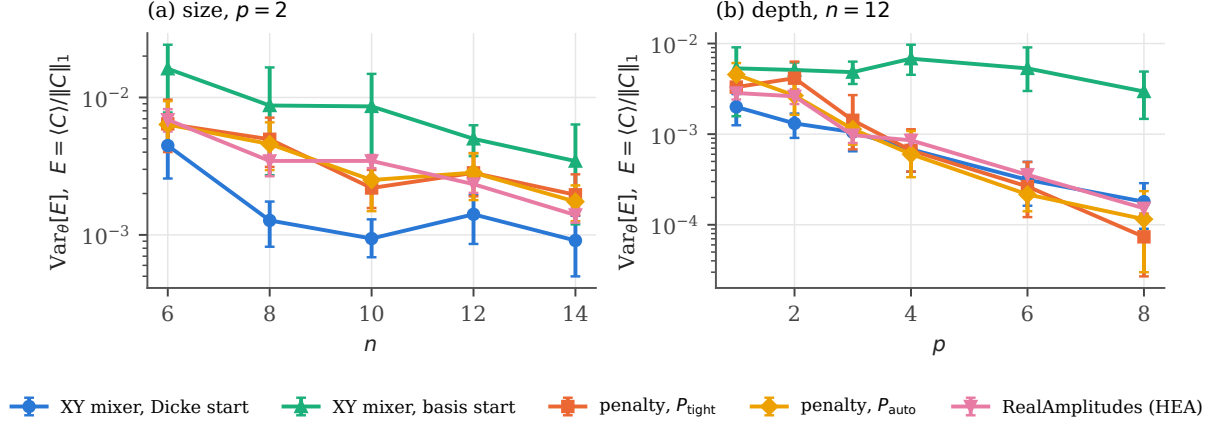


Figure 2: Variance of the rescaled cost over uniformly random parameters (5 instances $\times$ 200 samples per point, 95% CI). (a) Size scan at $p = 2$; (b) depth scan at $n = 12$ (120 samples).

Table 2: Fitted slope of $\ln$ variance per qubit, $n = 6\text{--}14$, $p = 2$ (95% bootstrap CI). For reference, a unitary 2-design gives $-\ln 2 \approx -0.69$.

Ansatz	$\text{Var}_\theta[E]$	$\overline{\text{Var}}_\theta[\partial_j E]$
penalty, P_{auto}	$-0.15 [-0.22, -0.09]$	$0.22 [0.15, 0.29]$
penalty, P_{tight}	$-0.15 [-0.21, -0.08]$	$0.06 [-0.05, 0.15]$
XY, basis start	$-0.18 [-0.29, -0.08]$	$-0.16 [-0.25, -0.09]$
XY, Dicke start	$-0.15 [-0.23, -0.09]$	$-0.10 [-0.20, -0.02]$
RealAmplitudes (HEA)	$-0.18 [-0.20, -0.15]$	$-0.30 [-0.31, -0.28]$

Parameters are drawn uniformly over full periods of the un-rescaled circuit: $\gamma \in [0, 2\pi)$, since all costs have integer spectra; $\beta \in [0, \pi)$ for the transverse mixer; $[0, 2\pi)$ for XY hops and R_Y angles. The measured observable is rescaled, $E = \langle C \rangle / \|C\|_1$ with $\|C\|_1$ the total Pauli weight, so that $|E| \leq 1$ for every method; the circuit is not rescaled. I report both the cost variance and the mean variance of the exact gradient components.

The cost variance decreases with n at the same rate for every ansatz, within overlapping intervals (Fig. 2a, Table 2). The rate is several times slower than the 2-design value, so none of these shallow circuits is near a barren plateau at these sizes. Over $p = 1 \rightarrow 8$ the cost variance falls by a factor of 10–45 for penalty QAOA, the Dicke-start XY mixer and the hardware-efficient ansatz. For the basis-start XY mixer it stays nearly flat, consistent with its amplitude remaining near the initial state (Fig. 2b).

The gradient variance agrees for the XY mixer and the HEA. For penalty QAOA it does not, and it even grows with n for P_{auto} . This is a matter of units: $\partial E / \partial \gamma$ carries a factor of the generator, whose spectrum is set by P , and P grows with n . Rescaling the generator inside the circuit would remove the factor but change the sampled region. The cost variance does not depend on the parametrisation, and exponential cost concentration and exponentially vanishing gradients imply each other [4], so it is the appropriate diagnostic. The conclusion for the research question is that, at these sizes, trainability does not explain the quality gap.

4.3 Gate cost after routing

In the idealised all-to-all circuits the XY ansatz is two to three times deeper than penalty QAOA. After routing, the order reverses (Table 3): the penalty term $P(\sum_i x_i - k)^2$ contains a $Z_i Z_j$ term for every pair, which

Table 3: Two-qubit (CZ) gates after transpilation for FakeMarrakesh (heavy-hex, optimisation level 1, mean over 5 transpiler seeds): penalty / XY (ratio).

n	$p = 1$	$p = 2$	$p = 3$
8	123 / 52 (2.34)	249 / 112 (2.23)	379 / 179 (2.11)
10	209 / 133 (1.57)	424 / 294 (1.44)	634 / 464 (1.37)
12	310 / 137 (2.27)	625 / 290 (2.15)	948 / 454 (2.09)
14	429 / 205 (2.09)	875 / 466 (1.88)	1328 / 717 (1.85)

a heavy-hex device implements only with many SWAPs, while the ring mixer and the sparse overlap terms route cheaply. Penalty QAOA needs 1.4–2.3 times the two-qubit gates; the ratio is lowest for the $n = 10$ instance, whose overlap graph is densest. Standard deviations over seeds are a few percent. Routed results depend on the number of CPU cores, because Qiskit’s Sabre pass runs one trial per core by default.

4.4 Re-analysis of a hardware run

Table 4: Job on ibm_marrakesh (4000 shots per circuit, $n = 12$, optimum 13). “Best of top 10”: best coverage among the 10 most frequent feasible bitstrings. Simulator column: same statistic on 4000 shots from the exact noiseless distribution (mean, 95% range). $P(U \geq \text{hw})$: probability that uniform bitstrings score at least the hardware value. The last two columns are the emulated feasible fraction (Aer with device calibration, 1000 shots), raw and readout-mitigated.

Circuit	hardware	simulator	$P(U \geq \text{hw})$	$E[f/f^* \text{feas.}]$, noiseless	emul. P_{feas}	mitigated
penalty, $p = 1$	13	12.3 [11, 13]	0.12	0.74	0.187	0.184
XY (basis), $p = 1$	9	6.0 [6, 6]	1.00	0.46	0.765	0.825
penalty, $p = 2$	12	10.9 [8, 12]	0.50	0.63	0.127	0.126
XY (basis), $p = 2$	10	8.0 [8, 8]	0.99	0.62	0.604	0.652
penalty, $p = 3$	11	11.3 [11, 12]	0.87	0.69	0.212	0.210
XY (basis), $p = 3$	11	11.8 [10, 12]	0.87	0.62	0.427	0.434

Six circuits (both encodings, $p = 1, 2, 3$, basis start for the XY mixer, one COBYLA run each) were run once on ibm_marrakesh; the reported statistic was the best coverage among the 10 most frequent feasible bitstrings. The circuits were rebuilt offline: their idealised depths match the recorded ones exactly. Applied to uniformly random 12-bit strings, the same statistic averages 11.5/13 (Table 4), and every hardware value lies within or below that null range. The circuits were also weak in noiseless simulation: the XY circuits reach a feasible-shot ratio of only 0.46–0.62, below the random value of 0.65. In a noisy emulation the XY circuits keep 43–77% of shots feasible, against 12% for uniform strings, so they are not fully depolarised. Tensorred readout mitigation changes these fractions by at most six points: gate errors dominate. Raw counts were not saved, so mitigation could not be applied to the real data. The run is therefore not evidence for either encoding.

5 Discussion

The quality result has a simple reading. With P_{auto} the penalty coefficients are an order of magnitude larger than the objective’s, so the cost spectrum is dominated by feasibility. A shallow circuit and its optimiser then spend their resources raising the feasible fraction (0.43 at $p = 1$ to 0.55 at $p = 3$ for $n = 12$) and barely bias the feasible distribution. P_{tight} shrinks the penalty by a factor of about 5 and helps only marginally, which suggests that the difficulty lies in the transverse mixer exploring all 2^n strings, not only in the size of P . The XY mixer avoids both problems, at the price of a weight- k initial state. Its basis-state version inherits an arbitrary bias from station labelling, which the Dicke start removes.

Supplementary checks (in the accompanying notebook). Warm-starting penalty QAOA from the LP relaxation [6] raises the feasible-shot ratio at $p = 3$ from 0.683 to 0.834 (family R, where the LP is integral) and from 0.733 to 0.827 (family H, where LP rounding is wrong). For multi-angle QAOA [7] at $n = 12$, the cost variance of standard QAOA falls from 4.4×10^{-3} at $p = 1$ to 7.9×10^{-6} at $p = 45$, and the 90-parameter ma-QAOA is already at 3.3×10^{-6} at $p = 1$. NFT [8] outperforms COBYLA on hardware-efficient ansätze, where its single-frequency assumption holds exactly; a ridge-regression initialiser for penalty QAOA gives no gain.

Limitations. (i) Scale: $n \leq 14$; the fitted slopes are descriptive. (ii) Hardness: every instance is easy for enumeration, and even family H for simulated annealing. (iii) The objective is a proxy, exact only up to double overlaps. (iv) The Dicke-state preparation is not costed. (v) The hardware evidence is a single job without raw counts.

Next steps. Simulating the XY mixer in its $\binom{n}{k}$ -dimensional feasible subspace makes $n = 16$ –24 accessible and would test whether the quality gap and the variance trends persist. Instances with a real LP integrality gap, beyond enumeration, are needed for any comparison with classical solvers. A repeated hardware run should save raw counts, include an on-device uniform-superposition circuit as null model, measure readout calibration in the same batch, and use circuits that are already good in simulation.

6 Conclusion

For a cardinality-constrained coverage problem at $n \leq 14$ and $p \leq 3$, enforcing the constraint in the mixer gives clearly better single-shot solutions than a quadratic penalty, which at this depth is barely distinguishable from random sampling. Cost concentration is similar for both encodings, and the constraint-native circuit is about twice as cheap in two-qubit gates on heavy-hex hardware. Evaluating encodings with the most probable bitstring or with best-of-many-shots can reverse this conclusion.

Reproducibility and acknowledgements. All numbers come from a single seeded execution of the accompanying notebook (pinned environment; tables in `results/`). Parts of the code and of this text were drafted with the help of an AI assistant; the author reviewed and is responsible for all content.

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